

armcfgen.py - User Manual

This manual is in three parts:

1. Writing a CSUB -- what a CSUB is, exactly how BASIC passes arguments to it (integers, floats, strings, arrays and string arrays), the rules you must follow, and worked examples in both **merge** and **join** form.
2. Building a CSUB -- turning your C into a pasteable CSUB ... END CSUB block with armcfgen.py.
3. Appendix A -- The CallTable -- every firmware routine a CSUB can call.

If you would rather not write C at all, mmb2csub.py converts an MMBasic SUB or FUNCTION into a CSUB for you, using these same CallTable routines -- see docs/mmb2csub.md. This manual is the one to read when you want control over the C, or when what you need is not something MMBasic can express.

1.1 What a CSUB is

A CSUB is a block of compiled machine code embedded directly in a BASIC program as hexadecimal words:

```
CSUB myfunc INTEGER
00000000
B5704B08 0004681B 68006841 ...
END CSUB
```

When BASIC executes myfunc x, the interpreter jumps straight into that machine code. A CSUB therefore lets you write performance-critical or hardware-level work in C (or assembler) and call it from BASIC like any other subroutine.

Three consequences follow from "it's just embedded machine code", and they shape everything else in this manual:

- There is no linker. A CSUB cannot call firmware functions or C library functions by name. The **only** way it reaches anything in the firmware is through the CallTable -- a table of function pointers the firmware exposes (see §1.5 and Appendix A).
- It must be self-contained and position independent. The blob is pasted into flash wherever your program happens to live, so it can't depend on a fixed load address -- the firmware does no relocation. The build handles this for you, including read-only data (see §1.4).
- It targets Cortex-M0+. Code built for the M0+ instruction set runs unchanged on both the RP2040 (M0+) and the RP2350 (M33), so one blob works on every PicoMite board, unless it reads the BASIC variable table (g_vartbl) or the Option structure, whose layouts differ between the chips and variants (see §1.5).

A CSUB is a **subroutine**: it returns nothing usable to BASIC (its C return value is ignored). It communicates results by writing back through its arguments. (A CFUNCTION is the value-returning sibling; this manual concentrates on CSUBs, but the argument rules are identical.)

1.2 How arguments are passed -- the golden rule

Every argument is passed to the CSUB as a pointer.

BASIC never passes a value in a register -- it passes the **address** of the

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argument's storage. Your C function therefore always receives pointers, one per BASIC argument, in order:

```
long long myfunc(long long *a, double *b, unsigned char *c)
/*          ^ INTEGER      ^ FLOAT      ^ STRING          */
```

Because you get the address of the caller's storage, **writing through the pointer changes the BASIC variable in place** -- that is how a CSUB returns results. The C return type is always long long and is ignored.

Two limits: a CSUB may take at most 10 arguments, and array **bounds are not passed** (see §1.3.4).

1.2.1 Variables vs expressions

The in-place behaviour only works when the caller passes a variable:

```
myfunc x      ' x is a variable -> CSUB gets &x, can change it
myfunc 5      ' 5 is an expression -> CSUB gets a pointer to a throwaway
myfunc a+b    ' temporary; any change is lost
```

Always pass a variable for anything the CSUB writes to.

1.3 The five kinds of argument

1.3.1 Integers

A BASIC integer is a 64-bit signed value. You receive a pointer to it:

```
long long myfunc(long long *a)
{
    *a = *a * 2;          /* doubles the caller's integer */
    return 0;
}
```

```
x% = 21 : myfunc x% : Print x%      ' 42
```

*Note: 64-bit add/subtract compile inline, but 64-bit multiply/divide need a helper that isn't in the blob (see §1.4). For *a * 2 GCC uses a shift, so it's fine; a general 64x64 multiply is not.*

1.3.2 Floats

A BASIC float is a C double (64-bit). You receive a pointer to it:

```
long long myfunc(double *a)
{
    *a = FMul(*a, *a);    /* squares it - note FMul, NOT * (see §1.4) */
    return 0;
}
```

1.3.3 Strings

A BASIC string is not a C string. It is a length-prefixed buffer:

```
byte 0      : length (0-255)
bytes 1..N  : the characters (NOT null-terminated)
```

You receive a pointer to byte 0:

```
long long supper(unsigned char *s) /* uppercase a string in place */
```

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```
{
    int n = s[0];                /* byte 0 is the length      */
    for (int i = 1; i <= n; i++) /* characters are s[1]..s[n] */
        if (s[i] >= 'a' && s[i] <= 'z') s[i] -= 32;
    return 0;
}
```

```
a$ = "hello" : supper a$ : Print a$      ' HELLO
```

To return a string, set the length byte and write the characters:

```
s[0] = 3; s[1] = 'a'; s[2] = 'b'; s[3] = 'c'; /* s$ becomes "abc" */
```

Maximum length is 255. If you want a normal null-terminated C string to work with, you must convert it yourself (read s[0], copy s[1..n], append \0); the firmware's internal CtoM/MtoC helpers are not in the CallTable.

1.3.4 Numeric arrays

An array argument (written a() in the call) passes a pointer to the array's contiguous data -- long long * for an integer array, double * for a float array:

```
long long isum(long long *a, long long *n, long long *result)
{
    long long s = 0;
    for (int i = 0; i < (int)*n; i++) s += a[i];
    *result = s;
    return 0;
}
```

```
Dim Integer v(4) = (10,20,30,40,50)
Dim Integer total
isum v(), 5, total
Print total      ' 150
```

Array bounds are not passed. The CSUB has no idea how big a() is, so pass the element count as a separate argument (as n above) -- or agree on a fixed size. Reading past the end corrupts memory.

You can also pass a single element (a(3)), which gives a pointer to that one element's storage.

1.3.5 String arrays

A string array passes a pointer to the first element. The elements are laid out back-to-back, each one a length-prefixed string of fixed width:

```
stride = (declared LENGTH) + 1      bytes per element
        = 256 by default              (LENGTH defaults to 255)
element e starts at  base + e * stride
```

So the CSUB needs both the element count and the stride (i.e. the LENGTH) to walk the array:

```
long long aupper(unsigned char *s, long long *nn, long long *len)
{
    int n = (int)*nn, stride = (int)*len + 1; /* LENGTH + 1 */
    for (int e = 0; e < n; e++) {
        unsigned char *el = s + e * stride; /* this element */
        int L = el[0];                      /* its length byte */
        for (int i = 1; i <= L; i++)
            if (el[i] >= 'a' && el[i] <= 'z') el[i] -= 32;
    }
}
```

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```
}  
    return 0;  
}
```

```
Dim s$(2) Length 16 = ("one","two","three")  
upper s$( ), 3, 16  
Print s$(0), s$(1), s$(2)      ' ONE TWO THREE
```

Pass the same LENGTH you declared the array with; if you used the default, the stride is 256.

1.3.6 Summary

BASIC argument	C parameter	Notes
integer `x%`	`long long *`	64-bit; write back to modify
float `x!`	`double *`	C `double`; do maths via CallTable (§1.4)
string `s\$`	`unsigned char *`	length-prefixed: `s[0]`=len, `s[1..]`=chars, max 255
integer array `a()`	`long long *`	contiguous; bounds not passed
float array `a()`	`double *`	contiguous; bounds not passed
string array `s\$()`	`unsigned char *`	stride = `LENGTH`+1 (default 256); each element length-prefixed
single element `a(i)`	pointer to that element	
expression `a+b`, literal	pointer to a temporary	read-only in effect

1.4 The rules you must follow

These are not style preferences -- break them and the blob crashes or won't build.

1. No floating-point or 64-bit multiply/divide/variable-shift C operators. Cortex-M0+ has no FPU, no 64-bit multiply and no divide, and the blob links without libgcc -- so those operators make the compiler emit a helper (`__aeabi_dadd`, `__aeabi_dmul`, `__aeabi_lmul`, `__aeabi_ldivmod`, `__aeabi_llsl`, ...) that is not in the blob. Do that maths by calling the firmware instead (§1.5).

For double, every operator needs a firmware call. For long long, only five do -- `*`, `/`, `%`, and `<</>>` by a variable count. Add, subtract, negate, compare, `&/|/^/~` and shifts by a constant all compile inline and are fine as written, as is all 32-bit integer maths.

From firmware 6.03.02b8 the five have CallTable slots of their own -- `LMul`, `LDiv`, `LMod`, `LShl`, `LAshr`, `LLsr` (§1.5) -- so 64-bit maths no longer means dropping to 32 bits. Note `LAshr` is the *arithmetic* right shift (sign propagating) and `LLsr` the *logical* one; `MMBasic`'s own `>>` operator is logical, so `LLsr` is the one that matches it.

***Safety net:** if you slip up, the build fails with `undefined reference to __aeabi_...` rather than producing a broken blob.

`memcpy`/`memset`/`memmove` are a special case. The compiler emits calls to these *by name*, without you writing one -- `char buf[64] = {0}`, a struct assignment or an array copy all become one -- so a CallTable macro cannot catch them and the symbols themselves have to be in the blob. Put `#define CSUB_MEM_SHIMS` before `#include "PicoCFunctions.h"` and you get shims that forward to the firmware's own routines, 20 bytes each. Leave it undefined and you save the space, at the price of an `undefined reference to memset` if the compiler wants one.

switch statements are fine: the tool compiles with `-fno-jump-tables`, so a

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switch (or an if-chain the optimiser would turn into one) becomes compares and branches rather than a jump table through a libgcc helper (`__gnu_thumb1_case_*`) that is not in the blob either.

2. Constant data (.rodata) is fine. const tables, strings and double literals work: the tool compiles position-independent (text-relative) code and appends the rodata to the blob, reached PC-relative -- so it runs wherever it lands in flash, with no firmware relocation. (Passing values in through an argument array is still slightly leaner when you have the choice.)

3. Pass a variable, not an expression, for anything you write back (§1.2.1).

4. At most 10 arguments. Bundle extras (and any state that must persist between calls) into a parameter array.

5. Target stays Cortex-M0+ so the blob runs on both chips (but see §1.5 if it reads `g_vartbl` or `Option`).

1.5 Calling firmware routines (the CallTable)

Because there's no linker, a CSUB reaches the firmware through the CallTable.

The easy way is to include the header and call routines by name:

```
#include "PicoCFunctions.h"
...
*a = FMul(*a, *b);          /* double multiply */
*a = FAdd(Sine(*a), *b);    /* sin(), add      */
DrawPixel(x, y, RGB_value); /* plot a pixel  */
```

The header locates the CallTable automatically at runtime (via the Cortex-M VTOR register), so there is no CallTable argument to pass and the same blob runs on every variant and both chips. Compile with `-I` pointing at the firmware tree so the header is found (Part 2).

The exception: the variable table and `Option`. The CallTable's routines behave the same everywhere, but two of the data structures it points to do not:

- An entry in the variable table (`g_vartbl`) holds an array's dimensions (`dims[]`) as five ints on the RP2350 and six shorts on the RP2040, so the size of an entry, and the offset of `dims` and every field after it, differ between the chips.
- The fields of the `Option` structure sit at different offsets on different variants (PicoMite, VGA, WEB, HDMI, ...) and chips.

A CSUB that reads either works only on the chip (and, for `Option`, the variant) it was compiled for. Build it with `PICORP2350` defined in `PicoCFunctions.h` for an RP2350 and commented out for an RP2040, with the variant selected at the top of the header. A CSUB that touches neither, which is almost every CSUB and everything `mmb2csub` generates, runs everywhere.

The routines you will use most:

Call	Purpose
<code>`FAdd(a,b)` `FSub(a,b)` `FMul(a,b)` `FDiv(a,b)`</code>	<code>`double` + - x /</code>
<code>`Sine(x)` `Cosine(x)` `Sqrt(x)` `Atan2(y,x)` `Power(b,e)`</code>	<code>`double` maths</code>
<code>`IntToFloat(i)` `FloatToInt(f)`</code>	convert (<code>FloatToInt</code> rounds)
<code>`LMul(a,b)` `LDiv(a,b)` `LMod(a,b)`</code>	<code>`long long` x / remainder</code>
<code>`LShl(a,n)` `LAsr(a,n)` `LLsr(a,n)`</code>	<code>`long long` shifts by a variable count</code>
<code>`DrawPixel(x,y,c)`</code>	plot one pixel (<code>c</code> is a full RGB colour)
<code>`Display_Refresh()`</code>	push direct drawing to a buffered panel
<code>`HRes` `VRes`</code>	current screen size

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DrawPixel is format- and version-independent: you pass a full RGB colour and the firmware packs it for whatever mode is active -- so it is the safe way to plot. The full list is in Appendix A.

There is also a single-precision (float) set -- SAdd/SSub/SMul/SDiv, SSin/SCos/SSqrt/SAtan2/SPow, and the conversions DtoS/StoD/StoI/ItoS. On the RP2350 these run on the M33 FPU and are much faster than the double routines, so they're worth it in hot maths (see Appendix A and bubble_s.c). Inputs from BASIC are double, so DtoS them to float first.

Hot loops: each header wrapper re-reads the table base. In a tight loop cache it once -- unsigned int base = BaseAddress; then use (base + 0x90) -- or, for maximum speed, write directly into the framebuffer via WriteBuf (Appendix A; note its bytes are mode-specific).

1.6 Worked examples

A. Square an integer -- no firmware calls, no header

Pure integer work reaches nothing in the firmware, so no header is needed.

```
long long sqr(long long *a)
{
    int v = (int)(*a);          /* 32-bit multiply, avoids __aeabi_lmul */
    *a = (long long)(v * v);
    return 0;
}
```

```
python armcfgen.py sqr.c --compile -n sqr -e sqr
```

```
x% = 5 : sqr x% : Print x%      ' 25
```

B. Square a double -- header + FMul (merge)

```
#include "PicoCFunctions.h"
long long sqrf(double *a)
{
    *a = FMul(*a, *a);
    return 0;
}
```

```
python armcfgen.py sqrf.c --compile -n sqrf -e sqrf -I path\to\PicoMite
```

```
x! = 1.5 : sqrf x! : Print x!   ' 2.25
```

C. Uppercase a string (merge)

The §1.3.3 supper example (no header -- pure byte work). Build it and call it:

```
python armcfgen.py supper.c --compile -n supper -e supper
```

```
a$ = "hello" : supper a$ : Print a$      ' HELLO
```

D. Several functions, one calling another (merge)

Merge mode puts every function in one CSUB so they can call each other:

```
__attribute__((noinline)) static int triple(int x){ return x + x + x; }
long long scale3(long long *a){ *a = (long long)triple((int)*a); return 0; }
```

```
python armcfgen.py scale3.c --compile -n scale3 -e scale3
```

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scale3 calls triple; both end up in a single CSUB block with the call between them resolved. The entry function (-e scale3) goes first.

E. A file of independent CSUBs (join)

Join mode emits a separate CSUB ... END CSUB for each function -- handy when you keep a library of unrelated CSUBs in one .c:

```
long long sqr(long long *a){ int v=(int)*a; *a=(long long)(v*v); return 0; }
long long neg(long long *a){ *a = -*a; return 0; }
```

```
python armcfgen.py lib.c --compile -m join
```

This produces a CSUB sqr ... block and a CSUB neg ... block, ready to paste individually. Functions in join mode cannot call each other (each is its own standalone blob).

F. Constant data -- .rodata (merge)

const data just works; the tool builds it position-independent:

```
#include "PicoCFunctions.h"
const char test[] = "12345678";
void main(void)
{
    MMPrintString((char *)test);
}
```

```
python armcfgen.py rotest.c --compile -n rotest -e main -I path\to\PicoMite
```

```
rotest          ' prints 12345678
```

The "12345678" string is appended to the blob and reached PC-relative, so it prints correctly wherever the CSUB lands in flash. Note main may be void -- the C return value is ignored either way.

armcfgen.py turns compiled ARM code into the CSUB ... END CSUB text. It can drive the compiler for you, so you go straight from .c to a pasteable block.

2.1 Prerequisites

- Python 3.8+ -- python --version
- pyelftools -- pip install pyelftools
- Arm GNU toolchain (arm-none-eabi-gcc) on your PATH -- the same one that builds the firmware. On Windows it is typically at C:\Program Files (x86)\Arm GNU Toolchain arm-none-eabi\<version>\bin.
- The header PicoCFunctions.h if your source #includes it. It ships in the firmware tree; point at it with -I rather than copying it.

(Only --compile and the link step use the toolchain; an already-linked .elf needs just Python + pyelftools.)

2.2 Quick start

```
python armcfgen.py sqrf.c --compile -n sqrf -e sqrf -I path\to\PicoMite
```

Add -o sqrf.bas to write to a file instead of the screen, then copy the CSUB ... END CSUB block into your program.

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path\to\PicoMite is wherever you unpacked the firmware source -- the folder holding PicoCFunctions.h. If you are running armcfgen.py from the user-tools folder it came in, that is simply -I ...

2.3 Command-line reference

```
python armcfgen.py INPUT [INPUT ...] [options]
```

Option	Default	Meaning
`INPUT`	--	A linked `.elf`, a relocatable `.o`, or `.c` source (with `--compile`). Several inputs are linked together.
`-c`, `--compile`	off	Compile the inputs as C first, with the built-in flags.
`-I`, `--include DIR`	--	Header search directory for `--compile` (the firmware tree, for `PicoCFunctions.h`). Repeatable.
`-m`, `--mode {join,merge}`	`merge`	Output style -- §2.4.
`-e`, `--entry NAME`	`main`	Entry symbol: sets the offset word in merge; dropped as a dummy in join.
`-n`, `--name NAME`	input stem, upper-cased	CSUB name in merge mode.
`-O`, `--opt LEVEL`	`0`	Optimisation level for `--compile`. `0` is the simple, reliable default; use `s` (or `2`) for compute-heavy CSUBs.
`-o`, `--output FILE`	stdout	Write to a file.

Built-in compile flags: `-mcpu=cortex-m0plus -mthumb -O0 -ffreestanding -fno-exceptions -fpie -mpic-data-is-text-relative -msingle-pic-base`. The text-relative PIC flags are what make the blob self-contained (rodata reached PC-relative, resolved at link -- no GOT, no firmware fixup). The list of functions found is printed to stderr.

2.4 join vs merge

merge (default) -- one CSUB containing every function in address order, with the entry-offset word pointing at --entry. Functions can call each other. Use for any single-function CSUB and for multi-function drivers.

```
python armcfgen.py driver.c helpers.c --compile -n driver -e main -I path\to\PicoMite
```

The entry function must be 4-byte aligned -- make it the first function (the offset word is word-granular; the tool errors if the entry lands mid-word).

join -- a separate CSUB name ... END CSUB per function, each with its own 00000000 offset; the entry symbol is dropped. Functions cannot call each other, and constant data is rejected.

```
python armcfgen.py lib.elf -m join
```

2.5 Automatic linking

If you pass more than one object, or any input is a relocatable .o, armcfgen.py links them into one resolved image (gathering .text.startup, .text* and .rodata* into a single .text at address 0). This is what makes merge with internal calls work: an un-linked .o still has the calls between functions stored as unresolved relocations, and GCC puts main in a separate .text.startup section. Linking fixes the calls and merges the fragments. A fully linked .elf is used as-is.

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2.6 Output, and the type list you must add

```
CSUB sqrf FLOAT
00000000
B5704B08 0004681B 68006841 ...
END CSUB
```

- line 1 -- CSUB, the name, and the argument type list you add yourself (the tool only knows the name). Types are INTEGER, FLOAT, STRING; arrays are written INTEGER, FLOAT, STRING too and passed as name().
- line 2 -- the entry-offset word (00000000 = entry at the first code word).
- then the machine code as little-endian 32-bit words, eight per line.

Match the type list to your call, e.g. a CSUB taking (double*, long long*) is declared CSUB myfunc FLOAT, INTEGER.

2.7 Using it in BASIC

Paste the block anywhere (conventionally at the end, like a SUB) and call it:

```
Dim Float x! = 3
sqrf x!
Print x!          ' 9
```

No CallTable argument is needed -- the blob finds the CallTable itself.

2.8 Troubleshooting

Message	Cause / fix
`arm-none-eabi-gcc: not found`	Toolchain not on `PATH` (§2.1).
`No module named 'elftools`	`pip install pyelftools`.
`PicoCFunctions.h: No such file`	Add `-I` pointing at the firmware tree.
`... is not fully linked. Re-run with --compile...`	An un-linked `.o` with unresolved relocations -- use `--compile` or a linked `.elf`.
`undefined reference to ' aeabi ...`	A `double`/64-bit operator slipped in -- route the maths through the CallTable (§1.4).
`undefined reference to ' _gnu_thumb1_case ...`	A jump table got in (an object built without the tool's flags) -- compile with `--compile`, or add `--fno-jump-tables` to your own build.
`entry '...' is not 4-byte aligned`	Put the merge entry function first in the source.
`constant data (.rodata) not allowed in JOIN mode`	Remove `const` tables / big literals, or use merge.
`no symbol table`	The object was stripped -- rebuild without stripping.

Every routine and data pointer a CSUB can reach, with its slot offset from the table base. With PicoCFunctions.h you use the name directly; the offset matters only if you resolve the table by hand. Slots are append-only -- a given offset always means the same thing across firmware versions, so a compiled CSUB keeps working.

Console, formatting and program control

Name	Offset	Prototype / use
`uSec`	0x00	`void uSec(unsigned long long us)` -- busy-wait microseconds
`putConsole`	0x04	`void putConsole(int ch, int flush)` -- write a character to the console
`getConsole`	0x08	`int getConsole(void)` -- read a character, -1 if none waiting
`MMPrintString`	0x20	`void MMPrintString(char *s)` -- print a null-terminated C string
`IntToStr`	0x24	`void IntToStr(char *dst, long long n, unsigned int base)`

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Name	Offset	Prototype / use
`FloatToStr`	0x70	`void FloatToStr(char *dst, MMFLOAT f, int intDig, int decDig, char pad)`
`CheckAbort`	0x28	`void CheckAbort(void)` -- service the interpreter; honours Ctrl-C
`error`	0x58	`void error(char *msg)` -- abort the CSUB with a BASIC error (C string)
`SoftReset`	0x54	`void SoftReset(void)` -- restart the firmware
`RunBasicSub`	0x74	`void RunBasicSub(char *name)` -- call a BASIC SUB (name ends in **two** nulls)

Pins and PIO

Name	Offset	Prototype / use
`ExtCfg`	0x0C	`void ExtCfg(int pin, int cfg, int option)` -- configure a pin (`EXT_DIG IN`, ...)
`ExtSet`	0x10	`void ExtSet(int pin, int val)` -- drive a digital output
`ExtInp`	0x14	`int ExtInp(int pin)` -- read a configured pin
`PinSetBit`	0x18	`void PinSetBit(int pin, unsigned int offset)` -- low-level pin register write
`PinRead`	0x1C	`int PinRead(int pin)` -- read a pin level
`ExtCurrentConfig`	0x48	`int ExtCurrentConfig[]` -- current configuration of each pin
`PIOExecute`	0xD8	`void PIOExecute(int pio, int sm, uint32_t instruction)`

Memory

Name	Offset	Prototype / use
`GetMemory`	0x2C	`void *GetMemory(size_t n)` -- allocate from the MMBasic heap (persists)
`GetTempMemory`	0x30	`void *GetTempMemory(int n)` -- temporary allocation (auto-freed)
`FreeMemory`	0x34	`void FreeMemory(void *p)` -- free a `GetMemory` block
`ProgFlash`	0x5C	`int *` -- base of program memory
`CFuncRam`	0x7C	`int *` -- persistent static RAM reserved for CFunctions

Maths

Name	Offset	Prototype / use
`IntToFloat`	0x84	`MMFLOAT IntToFloat(long long)`
`FloatToInt`	0x88	`long long FloatToInt(MMFLOAT)` -- rounds (x>=0 ? +0.5 : -0.5)
`Sine`	0x90	`MMFLOAT Sine(MMFLOAT)` -- sin
`FMul`	0xA0	`MMFLOAT FMul(MMFLOAT,MMMFLOAT)` -- a*x
`FAdd`	0xA4	`MMFLOAT FAdd(MMFLOAT,MMMFLOAT)` -- a+b
`FSub`	0xA8	`MMFLOAT FSub(MMFLOAT,MMMFLOAT)` -- a-b
`FDiv`	0xAC	`MMFLOAT FDiv(MMFLOAT,MMMFLOAT)` -- a/b
`FCmp`	0xB0	`int FCmp(MMFLOAT,MMMFLOAT)` -- -1 / 0 / 1
`LoadFloat`	0xB4	`MMFLOAT LoadFloat(unsigned long long bits)` -- reinterpret a bit pattern
`IDiv`	0xC8	`int IDiv(int a, int b)` -- 32-bit integer divide
`IMod`	0x13C	`int IMod(int a, int b)` -- 32-bit integer remainder
`LMul`	0x140	`long long LMul(long long, long long)` -- a x b
`LDiv`	0x144	`long long LDiv(long long, long long)` -- a / b (MMBasic `\\`)
`LMod`	0x148	`long long LMod(long long, long long)` -- a % b (MMBasic `MOD`)
`LShl`	0x14C	`long long LShl(long long, int n)` -- a << n
`LAsr`	0x150	`long long LAsr(long long, int n)` -- a >> n, **arithmetic** (sign propagating)
`LLsr`	0x154	`unsigned long long LLsr(unsigned long long, int n)` -- a >> n, **logical** ; this is what MMBasic's `>>` does
`memcpy`	0x158	`void *memcpy(void *, const void *, size_t)`
`memset`	0x15C	`void *memset(void *, int, size_t)`
`memmove`	0x160	`void *memmove(void *, const void *, size_t)`
`Log`	0x1A0	`MMFLOAT log(MMFLOAT)`
`Tan`	0x1A4	`MMFLOAT tan(MMFLOAT)`

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Name	Offset	Prototype / use
`RndVal`	0x19C	`MMFLOAT RndVal(void)` -- `RND`, same generator as BASIC
`TimerVal`	0x198	`MMFLOAT TimerVal(void)` -- `TIMER`, milliseconds
`TimerSet`	0x1AC	`void TimerSet(long long ms)` -- `TIMER = n`
`Cosine`	0xF4	`MMFLOAT cos(MMFLOAT)`
`Sqrt`	0xF8	`MMFLOAT sqrt(MMFLOAT)`
`Atan2`	0xFC	`MMFLOAT atan2(MMFLOAT y, MMFLOAT x)`
`Power`	0x100	`MMFLOAT pow(MMFLOAT base, MMFLOAT exp)`

Single precision (float)

MMFLOAT is a double; these routines operate on C float. On the RP2350 (firmware built pico_vfp) float runs on the M33 FPU, so single precision is markedly faster than double -- worth using in hot CSUB maths. BASIC numbers are still double, so convert at the boundary: DtoS the incoming pf() values to float, compute in float, then StoD/Stol to write results back (see bubble_s.c).

Name	Offset	Prototype / use
`SAdd`	0x104	`float SAdd(float,float)` -- a+b
`SSub`	0x108	`float SSub(float,float)` -- a-b
`SMul`	0x10C	`float SMul(float,float)` -- axb
`SDiv`	0x110	`float SDiv(float,float)` -- a/b
`SCmp`	0x114	`int SCmp(float,float)` -- -1 / 0 / 1
`SSin`	0x118	`float SSin(float)` -- sinf
`SCos`	0x11C	`float SCos(float)` -- cosf
`SSqrt`	0x120	`float SSqrt(float)` -- sqrtf
`SAtan2`	0x124	`float SAtan2(float y, float x)` -- atan2f
`SPow`	0x128	`float SPow(float base, float exp)` -- powf
`DtoS`	0x12C	`float DtoS(double)` -- double -> single
`StoD`	0x130	`double StoD(float)` -- single -> double
`Stol`	0x134	`long long Stol(float)` -- single -> int (rounds)
`ltoS`	0x138	`float ltoS(long long)` -- int -> single

*On the RP2040 (no FPU) these are *software* single precision -- still faster than software double, but not FPU-accelerated. The double routines above use the DCP coprocessor on RP2350.*

Pins

SETPIN and the rest of the configuration belong in BASIC, where they happen once. What a CSUB is for is watching or driving a pin in a tight loop, and these are the interpreter's own routines behind PIN(n) and PIN(n) = v, so a CSUB and BASIC cannot read a pin differently.

PinVal returns whatever the pin's mode means -- a digital level, a raw ADC count, a filtered voltage, a frequency or a period -- exactly as PIN(n) does. The pin number is the one PIN() takes, which for a GPnn is MM.INFO(PINNO GPnn).

Neither can report an error, because a routine reachable from a CSUB must not call error(). An invalid pin, a pin that is not an input, or an ADC busy with a DMA transfer reads 0 rather than stopping the program. PIN() in BASIC still raises all of those.

Name	Offset	Prototype / use
`PinVal`	0x1B8	`MMFLOAT PinVal(int pin)` -- `PIN(n)`

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Name	Offset	Prototype / use
`PinPut`	0x1BC	`void PinPut(int pin, int val)` -- `PIN(n) = v`
`ExtCurrentConfig`	0x1C0	`volatile int *` -- what `SETPIN` made each pin (`EXT_DIG_IN`, `EXT_ANA_IN` ...), indexed by pin number

The lower-level ExtSet, ExtInp, ExtCfg, PinRead and PinSetBit are also available (see Console, formatting and program control above) and bypass the mode dispatch; use PinVal unless you know exactly what the pin is.

PinSetBit is the fast path, and it is worth knowing about. PIN(n) = v goes through ExtSet, which checks the pin's mode and then calls PinSetBit(LATSET) or PinSetBit(LATCLR) -- and each of *those* re-establishes the pull configuration before writing the level, three SDK calls a write. PinSetBit(pin, LATINV) is one gpio_xor_mask64 and nothing else. There is no MMBasic statement for it, because PIN(n) = v has to name a level, so it is reachable only from a CSUB.

Measured on an RP2040 at 315 MHz, toggling one DOUT pin (Bas/pinbench.bas, which times an empty loop too so the pin cost is separated from the loop's):

	ns per transition	cycles	rate
interpreted `PIN(n) = v`	7135	2248	126 kHz
CSUB, `PIN(n) = v`	572	180	1.6 MHz
CSUB, `PinSetBit(pin, LATINV)`	134	42	**5.5 MHz**

53x over the interpreter, and 4.3x over the converted PIN(n) = v. Use it when the pin is a known DOUT and you are toggling rather than setting a level -- ExtSet's bookkeeping is what you are skipping, so skip it only when you know it does not matter.

MMBasic strings

Added in 6.03.02b9. An MMBasic string is a **length byte followed by the data**: s[0] is the length and s+1 the first character, so it is not a C string and strlen/strcpy do not apply to it.

MtoC and CtoM convert in place, between that form and a NUL-terminated C string, and each returns its argument.

Name	Offset	Prototype / use
`MtoC`	0x164	`unsigned char *MtoC(unsigned char *s)` -- MMBasic form to C string, in place
`CtoM`	0x168	`unsigned char *CtoM(unsigned char *s)` -- C string to MMBasic form, in place
`Mstrcpy`	0x16C	`void Mstrcpy(unsigned char *dst, unsigned char *src)`
`Mstrcat`	0x170	`void Mstrcat(unsigned char *dst, const unsigned char *src)`
`Mstrcmp`	0x174	`int Mstrcmp(const unsigned char *, const unsigned char *)`

These are the interpreter's own implementations of the string functions, so a CSUB and BASIC cannot disagree about what MID\$ means. Each takes the destination as its first argument and returns it; GetTempStrMemory (0x30) gives you a buffer of the right size.

They never call error(), because error() does a longjmp that would abandon your CSUB's stack frame. They clamp instead: an out-of-range start or count is silently brought into range rather than reported. Check the arguments yourself if you need MMBasic's error.

Name	Offset	Prototype / use
`StrLeft`	0x178	`unsigned char *StrLeft(unsigned char *dst, const unsigned char *s, int n)` -- `LEFT\$`

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Name	Offset	Prototype / use
`StrRight`	0x17C	`unsigned char *StrRight(unsigned char *dst, const unsigned char *s, int n)` -- `RIGHT\$`
`StrCase`	0x180	`unsigned char *StrCase(unsigned char *dst, const unsigned char *s, int upper)` -- `UCASE\$` when upper, `LCASE\$` when 0
`StrMid`	0x184	`unsigned char *StrMid(unsigned char *dst, const unsigned char *s, int spos, int nbr)` -- `MID\$`, spos is 1-based
`StrChar`	0x188	`unsigned char *StrChar(unsigned char *dst, int c)` -- `CHR\$`
`StrFill`	0x18C	`unsigned char *StrFill(unsigned char *dst, int ch, int n)` -- `SPACE\$` (ch=' ') and `STRING\$`
`StrInstr`	0x190	`int StrInstr(const unsigned char *s1, const unsigned char *s2, int start)` -- `INSTR`, 1-based, 0 if not found
`StrFormat`	0x194	`unsigned char *StrFormat(unsigned char *dst, MMFLOAT f, long long i64, int isint, int m, int n, int ch)` -- `STR\$`; pass `isint` to choose which of `f`/`i64` is read, and `STR_AUTO_PRECISION` for n to get BASIC's default
`StrVal`	0x1A8	`int StrVal(const unsigned char *p, MMFLOAT *f, long long *i)` -- `VAL`, on a C string; returns non-zero if the result is an integer

Graphics

Name	Offset	Prototype / use
`DrawPixel`	0xEC	`void DrawPixel(int x, int y, int rgb)` -- plot one pixel; firmware packs the colour for the active mode (format-independent)
`DrawLine`	0x40	`void DrawLine(int x1,int y1,int x2,int y2,int w,int c)`
`DrawRectangle`	0x38	`void DrawRectangle(int x1,int y1,int x2,int y2,int c)` -- filled (inclusive)
`DrawCircle`	0x94	`void DrawCircle(int x,int y,int r,int w,int c,int fill,MMFLOAT aspect)`
`DrawTriangle`	0x98	`void DrawTriangle(int x0,int y0,int x1,int y1,int x2,int y2,int c,int fill)`
`DrawBitmap`	0x3C	`void DrawBitmap(int x,int y,int w,int h,int scale,int fg,int bg,unsigned char *bmp)`
`DrawBuffer`	0x68	`void DrawBuffer(int x1,int y1,int x2,int y2,char *src)` -- blit a pixel block in
`ReadBuffer`	0x6C	`void ReadBuffer(int x1,int y1,int x2,int y2,char *dst)` -- read a pixel block out
`ScrollLCD`	0x80	`void ScrollLCD(int lines, int blank)`
`FontTable`	0x44	`const unsigned char *FontTable[]` -- installed font data
`HRes`	0x4C	`unsigned int` -- current horizontal resolution
`VRes`	0x50	`unsigned int` -- current vertical resolution

Framebuffer (direct access)

Name	Offset	Prototype / use
`WriteBuf`	0xDC	`unsigned char *` -- current write framebuffer. Bytes are mode-specific (RGB121 = 4 bpp / 2 px-per-byte; RGB332 = 1 byte/px), so use only when you know your mode. Prefer `DrawPixel` otherwise.
`FrameBuf`	0xE0	`unsigned char *` -- frame layer
`LayerBuf`	0xE4	`unsigned char *` -- overlay layer
`DisplayBuf`	0xE8	`unsigned char *` -- display layer
`Display_Refresh`	0xF0	`void Display_Refresh(void)` -- push direct writes to a buffered/SPI panel (no-op on DMA-scanned VGA/HDMI)

Audio

Name	Offset	Prototype / use
`AudioOutput`	0xC4	`void AudioOutput(uint16_t left, uint16_t right)`
`AUDIO_WRAP`	0xCC	`uint16_t` -- current audio PWM wrap (full-scale) value

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System data and interrupt vectors

Name	Offset	Prototype / use
`Option`	0x8C	`struct option_s *` -- the firmware option/settings structure; its layout depends on the variant and chip (§1.5)
`g_vartbl`	0x60	the BASIC variable table; its entry layout differs between RP2040 and RP2350 (§1.5)
`g_varcnt`	0x64	number of variables in the table
`Timer`	0x9C	`unsigned long long Timer(void)` -- microsecond timer
`CFuncmSec`	0x78	vector slot for a millisecond-tick CFunction
`CFuncInt1`...`CFuncInt4`	0xB8,0xBC,0xD0,0xD4	vector slots for interrupt CFunctions
`CSubComplete` / `Interrupt`	0xC0	`char` -- interrupt-in-progress flag

Slots 0x84+ assume current firmware. The framebuffer/maths group at 0xDC-0x100 was added later; older firmware stops at PIOExecute (0xD8).